ISTANBUL TECHNICAL UNIVERSITY ★ GRADUATE SCHOOL

DESIGN OF A COMPACT BIDIRECTIONAL RF FRONT-END MODULE FOR 24–30 GHz 5G APPLICATIONS

M.Sc. THESIS

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İSTANBUL TEKNİK ÜNİVERSİTESİ ★ LİSANSÜSTÜ EĞİTİM ENSTİTÜSÜ

24–30 GHz 5G UYGULAMALARI İÇİN KOMPAKT ÇİFT YÖNLÜ RF ÖN UÇ MODÜLÜ TASARIMI

YÜKSEK LİSANS TEZİ

Hüseyin Cahit ERBİL
(504231213)

Elektronik ve Haberleşme Mühendisliği Anabilim Dalı

Elektronik Mühendisliği Programı

Tez Danışmanı: Doç. Dr. Nil Banu TARIM

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Hüseyin Cahit ERBİL, a M.Sc. student of ITU Graduate School student ID 504231213 successfully defended the thesis entitled “DESIGN OF A COMPACT BIDIRECTIONAL RF FRONT-END MODULE FOR 24–30 GHz 5G APPLICATIONS”, which he/she prepared after fulfilling the requirements specified in the associated legislations, before the jury whose signatures are below.

Thesis Advisor :	**Doç. Dr. Nil Banu TARIM** Istanbul Technical University	....
Jury Members :	**Prof. Dr. Name SURNAME** University	....
	Prof. Dr. Name SURNAME University	....
		....

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FOREWORD

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June 2026 Hüseyin Cahit ERBİL

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ABBREVIATIONS

5G	Fifth Generation
AiP	Antenna-in-Package
CMOS	Complementary Metal-Oxide-Semiconductor
EIRP	Equivalent Isotropically Radiated Power
FEM	Front-End Module
FR2	Frequency Range 2
LNA	Low-Noise Amplifier
MIMO	Multiple-Input Multiple-Output
NF	Noise Figure
OP1dB	Output 1-dB Compression Point
PA	Power Amplifier
PAE	Power-Added Efficiency
RFIC	Radio-Frequency Integrated Circuit
RX	Receiver
TDD	Time-Division Duplexing
TX	Transmitter

SYMBOLS

A	Physical antenna aperture
d	Antenna element spacing
f	Frequency
G	Antenna gain
N	Number of antenna elements
$P_{out,ch}$	Output power per channel
λ	Wavelength
θ_{BW}	Approximate beamwidth

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DESIGN OF A COMPACT BIDIRECTIONAL RF FRONT-END MODULE FOR 24–30 GHz 5G APPLICATIONS

SUMMARY

This thesis presents the design of an ultra-compact bidirectional Ka-band RF front-end module implemented in 65-nm CMOS technology for 5G FR2 applications covering the 24–30 GHz frequency range. Millimeter-wave phased-array and massive MIMO systems require a large number of repeated RF front-end channels. Therefore, the silicon area and power consumption of each channel become critical design constraints in addition to conventional RF metrics such as gain, noise figure, linearity, output power, and stability. Conventional front-end implementations generally employ separate low-noise amplifier (LNA), power amplifier (PA), and matching circuits, which increases the area overhead and limits scalability in highly integrated multi-channel arrays. This thesis addresses this problem by investigating a bidirectional front-end architecture in which the transmitter and receiver paths share the same active and passive signal chain in a half-duplex manner.

The proposed architecture consists of three cascaded bidirectional active gain stages interconnected by transformer-based matching circuits. Four transformer-based passive matching circuits are employed at the input, output, and interstage interfaces to provide broadband impedance transformation, differential signal conversion, and compact matching over the target band. The first stage is intentionally designed as an asymmetric bidirectional gain cell. In transmitter mode, the antenna-side device employs a cascode configuration to improve voltage handling capability and output swing. In receiver mode, the corresponding receive path avoids unnecessary stacking in order to reduce parasitic loading and preserve noise performance. The second and third stages employ symmetric cross-coupled bidirectional gain cells. Their operation direction is controlled by tail-current steering rather than supply switching, allowing the gate bias voltages of the core transistors to remain fixed and helping the devices remain within the safe operating area.

Capacitive neutralization is used as an important design technique throughout the front-end. In the first stage, the parasitic capacitances of the inactive PA path contribute to the neutralization of the LNA mode without introducing a significant additional area penalty. In the symmetric second and third stages, the inherent gate–drain capacitances of the cross-coupled devices provide effective neutralization, improving reverse isolation, stability, and the achievable maximum gain. The transformer-based matching circuits are designed using the top metal layers of the CMOS process and are verified through electromagnetic simulations. These passive structures are essential for achieving broadband operation while maintaining a compact layout.

The system-level motivation is also discussed from the perspective of 5G FR2 beamforming arrays. The relationship between channel output power, antenna element gain, number of array elements, and effective isotropic radiated power (EIRP) is used to connect the circuit-level performance to array-level requirements. For example, a 16-channel coherent array with approximately 10 dBm output compression power per channel and 5 dBi antenna element gain can provide an estimated EIRP of about 39 dBm before modulation back-off. This analysis clarifies the relevance of a compact 10-dBm-class front-end channel for user equipment, customer-premises equipment, and small-cell-class phased-array applications.

The simulated results show that the proposed front-end covers the full 24–30 GHz 5G FR2 lower spectrum. In the lower LNA band of 24.25–27.5 GHz, the receiver gain is approximately 19–24 dB, with simulated NFmin of 4.2–4.8 dB and NF of about 4.4–5.5 dB. In the upper LNA band of 26.5–29.5 GHz, the receiver gain is approximately 17.5–20 dB, with simulated NFmin of 4.4–5.0 dB and NF of about 4.7–5.3 dB. In transmitter mode, the front-end provides a peak gain of approximately 28 dB and maintains broadband operation across 24–30 GHz. The simulated output 1-dB compression point reaches approximately 10 dBm around 25–26 GHz, while the peak saturated output power is approximately 13 dBm. The TX and RX DC power consumptions are approximately 196 mW and 96 mW, respectively, and the core area is approximately 0.2 mm^2. These results demonstrate that the proposed architecture provides a compact and scalable solution for 5G FR2 bidirectional front-end and beamformer-channel applications.

Keywords: 5G FR2, Ka-band, CMOS RFIC, bidirectional front-end, beamforming, phased array, power amplifier, low-noise amplifier.

24–30 GHz 5G UYGULAMALARI İÇİN KOMPAKT ÇİFT YÖNLÜ RF ÖN UÇ MODÜLÜ TASARIMI

ÖZET

Bu tezde, 24–30 GHz frekans aralığını kapsayan 5G FR2 uygulamaları için 65 nm CMOS teknolojisinde tasarlanan kompakt çift yönlü bir Ka-band RF ön uç modülü sunulmaktadır. Milimetre-dalga faz dizili sistemlerde ve massive MIMO mimarilerinde çok sayıda RF ön uç kanalının aynı tümdevre veya paket içerisinde tekrarlanması gerekir. Bu nedenle tek bir kanalın kazanç, gürültü sayısı, doğrusallık ve çıkış gücü gibi klasik RF başarım ölçütleri; çip alanı, güç tüketimi, pasif eleman alanı ve termal yoğunluk gibi entegrasyon odaklı kısıtlarla birlikte değerlendirilmelidir. Geleneksel ön uç yapılarında düşük gürültülü kuvvetlendirici (LNA), güç kuvvetlendirici (PA) ve bunlara ait uyumlama devreleri ayrı verici ve alıcı yolları üzerinde gerçeklenir. Bu yaklaşım blokların bağımsız olarak optimize edilmesini kolaylaştırsa da, özellikle çok kanallı beamformer tümdevrelerinde tekrarlanan aktif cihazlar, transformatörler, balunlar ve yönlendirme hatları nedeniyle toplam alanı artırır. Bu tezde bu probleme karşılık, verici ve alıcı yollarının yarı çift yönlü çalışma prensibiyle aynı aktif ve pasif sinyal zincirini paylaştığı çift yönlü bir ön uç mimarisi ele alınmıştır.

Önerilen mimari, transformatör tabanlı uyumlama devreleri ile birbirine bağlanan üç çift yönlü aktif kazanç katından oluşmaktadır. Giriş, çıkış ve katlar arası arayüzlerde toplam dört adet transformatör tabanlı pasif yapı kullanılarak empedans dönüşümü, diferansiyel sinyal dönüşümü, DC kutuplama izolasyonu ve geniş bantlı çalışma aynı fiziksel yapılar üzerinden sağlanmıştır. İlk kazanç katı, verici ve alıcı modlarının farklı gereksinimlerini dengelemek amacıyla asimetrik olarak tasarlanmıştır. Verici modunda anten tarafındaki yapı daha yüksek gerilim salınımı ve çıkış gücü elde edebilmek için kaskot transistor yapısından yararlanırken, alıcı modunda gereksiz kaskotlamadan kaçınılarak parazitik yüklerin ve gürültü bozunumunun azaltılması hedeflenmiştir. İkinci ve üçüncü katlar ise simetrik çift yönlü kazanç hücreleri olarak tasarlanmıştır. Bu katlarda yön seçimi besleme geriliminin anahtarlanmasıyla değil, kuyruk akımının yönlendirilmesiyle yapılmaktadır. Böylece çekirdek transistorların kapı kutuplamaları sabit tutulmakta, anahtarlamadan kaynaklanan büyük parazitiklerin sinyal yolu üzerindeki etkisi azaltılmakta ve güvenli çalışma bölgesi açısından daha düzenli bir kutuplama ortamı elde edilmektedir.

Mimarinin temel tasarım unsurlarından biri kapasitif nötralizasyondur. Milimetre-dalga frekanslarında MOS transistorların kapı-savak kapasiteleri ters yönde kuplaj oluşturarak kazancı, kararlılığı ve izolasyonu sınırlayabilir. Önerilen yapıda bu etki yalnızca bastırılması gereken bir parazitik olarak değil, uygun şekilde kullanıldığında kazanç ve izolasyona katkı sağlayan bir tasarım aracı olarak

değerlendirilmiştir. İlk katta, verici yolunun alıcı modunda kapalı durumda bulunan parazitik kapasiteleri nötralizasyona yardımcı olacak şekilde kullanılmıştır. İkinci ve üçüncü katlarda ise simetrik çapraz bağlı yapı doğal bir nötralizasyon mekanizması oluşturmaktadır. Böylece ek büyük pasif eleman alanı gerektirmeden ters izolasyonun artırılması, maksimum erişilebilir kazancın iyileştirilmesi ve kararlılık marjının güçlendirilmesi hedeflenmiştir.

Tezde kullanılan transformatör tabanlı pasif yapılar yalnızca ideal devre elemanı olarak ele alınmamış, fiziksel yerleşim ve elektromanyetik etkilerle birlikte değerlendirilmiştir. Uyumlama devrelerinin hedef empedans dönüşümünü sağlayabilmesi için transformatör sarım oranı, manyetik kuplaj katsayısı, sargı endüktansları, metal kayıpları, parazitik kapasiteler ve öz rezonans frekansı birlikte dikkate alınmıştır. Üst metal katmanlarının kullanımı seri direnç kayıplarını azaltırken, kompakt ve yüksek kalite faktörlü pasif yapıların elde edilmesine katkı sağlamıştır. Bununla birlikte aynı transformatörlerin hem verici hem de alıcı modlarında çalışması, iki mod arasında eş zamanlı bir ödünleşim yapılmasını gerektirmiştir. Bu nedenle uyumlama devreleri, yalnızca tek bir çalışma modu için değil, çift yönlü sinyal yolu ve sistem seviyesi gereksinimler göz önünde bulundurularak tasarlanmıştır.

Sistem seviyesi değerlendirmede 5G FR2 beamforming dizileri için kanal başına çıkış gücü, anten eleman kazancı, dizi eleman sayısı ve eşdeğer izotropik yayılım gücü (EIRP) arasındaki ilişki incelenmiştir. Tek bir RF kanalından daha yüksek çıkış gücü elde etmek genellikle daha büyük transistor boyutları ve daha yüksek kutuplama akımları gerektirir. Bu durum çip alanını ve güç tüketimini artırdığı gibi sınırlı bir tümdevre alanı içinde güç yoğunluğunu ve termal yoğunluğu da yükseltir. Bu nedenle faz dizili sistemlerde ön uç kanalı yalnızca mutlak çıkış gücüyle değil, alan verimliliği, güç tüketimi, ısı dağılımı ve dizi ölçeğinde tekrarlanabilirlik açısından da değerlendirilmelidir. Bu tezde önerilen ön uç, tek başına yüksek güçlü bir güç kuvvetlendirici olarak değil, çok kanallı bir beamformer tümdevresinde tekrarlanabilen kompakt bir anten tarafı RF kanalı olarak yorumlanmıştır.

Bu bakış açısıyla, kanal başına yaklaşık 10 dBm çıkış 1-dB sıkışma gücü ve 5 dBi anten eleman kazancı varsayıldığında, 16 kanallı koherent bir dizinin modülasyon geri çekmesi uygulanmadan önce yaklaşık 39 dBm EIRP sağlayabileceği gösterilmiştir. Dizi eleman sayısı arttıkça koherent birleşim kazancı yükselmekte ve her bir kanaldan beklenen çıkış gücü gereksinimi gevşemektedir. Bu nedenle önerilen 10-dBm sınıfı kompakt ön uç kanalı; kullanıcı ekipmanı, sabit kablosuz erişim terminalleri, küçük hücre sınıfı uygulamalar ve alan kısıtlı çok kanallı beamformer yapıları için anlamlı bir çözüm olarak değerlendirilebilir. Bu değerlendirme, devre seviyesi başarımın sistem seviyesi hedeflerle birlikte ele alınmasının önemini göstermektedir.

Benzetim sonuçları, önerilen ön ucun 24–30 GHz aralığındaki 5G FR2 alt spektrumunu kapsadığını göstermektedir. 24.25–27.5 GHz alt LNA bandında alıcı kazancı yaklaşık 19–24 dB aralığında elde edilmiştir. Bu bantta minimum gürültü sayısı yaklaşık 4.2–4.8 dB, gürültü sayısı ise yaklaşık 4.4–5.5 dB seviyesindedir. 26.5–29.5 GHz üst LNA bandında alıcı kazancı yaklaşık 17.5–20 dB aralığında olup, minimum gürültü sayısı yaklaşık 4.4–5.0 dB ve gürültü sayısı yaklaşık 4.7–5.3 dB olarak bulunmuştur. Verici modunda ön uç 24–30 GHz boyunca geniş bantlı kazanç sağlamakta ve yaklaşık 28 dB tepe kazanç sunmaktadır. Çıkış 1-dB sıkışma noktası

25–26 GHz civarında yaklaşık 10 dBm seviyesine ulaşmakta, tepe doyum çıkış gücü ise yaklaşık 13 dBm olarak elde edilmektedir.

Güç tüketimi ve alan sonuçları da mimarinin hedeflenen kullanım bağlamı açısından önemlidir. Alıcı modunda yaklaşık 96 mW, verici modunda ise yaklaşık 196 mW DC güç tüketimi elde edilmiştir. Çekirdek alan yaklaşık 0.2 mm^2 mertebesindedir. Bu değerler, önerilen mimarinin yüksek derecede alan paylaşımı sağlayan bir ön uç yapısı olduğunu göstermektedir. Alanın küçük tutulması yalnızca tek kanal için değil, çok kanallı dizilerde toplam çip alanı, paket karmaşıklığı, pasif yönlendirme alanı ve üretim maliyeti açısından da önemlidir. Bu çalışmanın bir diğer katkısı, önerilen çift yönlü ön uç mimarisinin standart 65 nm CMOS teknolojisinde gerçekleştirilebilirliğinin alan, güç tüketimi, pasif entegrasyon ve başarım ölçütleri açısından araştırılmasıdır.

Sonuç olarak bu tez, 24–30 GHz 5G FR2 uygulamaları için kompakt, transformatör tabanlı ve çift yönlü bir RF ön uç mimarisi önermektedir. Önerilen yapı, verici ve alıcı sinyal yollarını ortak aktif katlar ve ortak pasif uyumlama devreleri üzerinden birleştirerek alan verimliliğini artırmayı hedeflemektedir. Asimetrik ilk kat, simetrik çift yönlü ikinci ve üçüncü katlar, kuyruk akımı tabanlı yön seçimi ve kapasitif nötralizasyon teknikleri birlikte kullanılarak kazanç, gürültü sayısı, çıkış gücü, izolasyon ve kararlılık arasında dengeli bir çözüm elde edilmiştir. Çalışmanın sonraki aşamalarında tam çip elektromanyetik doğrulama, üretim sonrası ölçümler, paket ve anten arayüzü etkileri ile proses, gerilim ve sıcaklık değişimlerinin ayrıntılı olarak incelenmesi gerekmektedir.

Anahtar Kelimeler: 5G FR2, Ka-band, CMOS RFIC, çift yönlü ön uç, beamforming, faz dizisi, güç kuvvetlendirici, düşük gürültülü kuvvetlendirici.

1. INTRODUCTION

1.1 Motivation

The rapid growth of mobile data traffic has motivated the use of millimeter-wave frequency bands in fifth-generation wireless communication systems. In particular, frequency range 2 (FR2) enables wide channel bandwidths around the 24–30 GHz range. However, operation at millimeter-wave frequencies introduces severe propagation loss, increased circuit sensitivity to parasitics, and strict requirements on phased-array front-end integration.

Large-scale multiple-input multiple-output (MIMO) arrays are commonly employed to compensate for the increased free-space path loss. This approach requires many parallel RF front-end channels, making the silicon area and power consumption of each channel critical design parameters. Recent bidirectional beamformer and front-end studies show that sharing TX/RX circuitry is an effective way to reduce the area of each array pixel and to improve the scalability of dense phased-array implementations [3]–[5]. Therefore, compact front-end architectures that can share circuit resources between transmitter and receiver modes are highly attractive for scalable phased-array systems.

1.2 Challenges in Millimeter-Wave Front-End Design

Millimeter-wave front-end circuits must simultaneously satisfy gain, noise figure, output power, linearity, stability, and area requirements. In conventional implementations, the low-noise amplifier (LNA) and the power amplifier (PA) are designed as separate blocks. Although this approach simplifies independent optimization, it increases the silicon area due to duplicated active devices, matching circuits, and routing structures.

At 24–30 GHz, passive components such as transformers, inductors, baluns, and transmission lines occupy a considerable portion of the total chip area. As the number of antenna elements increases, the area overhead associated with repeated matching circuits, routing structures, and gain stages becomes a limiting factor. Therefore, the front-end architecture must provide sufficient transmitter output power while maintaining low receiver noise figure within a compact silicon footprint.

1.3 Bidirectional Front-End Concept

A bidirectional front-end reuses the same physical signal path for transmitter and receiver operation in a half-duplex manner. This concept is particularly suitable for time-division duplexing systems, where the transmitter and receiver are not active simultaneously. By sharing active devices and passive matching structures, a bidirectional architecture can reduce the area per array element.

The main challenge is that the optimum design conditions for transmitter and receiver modes are generally different. A PA requires high output swing, proper load impedance, and sufficient device reliability margin, whereas an LNA requires low input-referred noise, high gain, and proper source impedance. The proposed design addresses this conflict by using an asymmetric first stage and shared bidirectional gain stages.

1.4 Contributions of the Thesis

The main contributions of this thesis are summarized as follows:

- Design of a compact bidirectional RF front-end module covering the 24–30 GHz 5G FR2 frequency range.
- Development of an asymmetric first-stage architecture to balance transmitter output capability and receiver noise performance.
- Implementation of current-steering based bidirectional operation without supply switching.
- Design of a three-stage transformer-coupled shared signal path for compact TX/RX hardware reuse.
- Application of capacitive neutralization techniques to improve gain, stability, and reverse isolation.
- Electromagnetic design and verification of transformer-based matching circuits for the input, output, and interstage interfaces.
- Investigation of the feasibility of implementing the proposed bidirectional front-end architecture in a standard 65-nm CMOS technology, including area, power-consumption, and integration trade-offs.
- System-level interpretation of the proposed front-end in terms of EIRP, OP1dB, noise figure, area, and scalability for multi-channel 5G beamformer ICs.

1.5 Organization of the Thesis

Chapter 2 presents the system-level and circuit-level background required to understand the proposed front-end. It introduces 5G FR2 phased-array operation, the beamformer channel concept, EIRP and output-power requirements, receiver noise-figure considerations, bidirectional front-end topologies, transformer-based matching methodology, neutralization, technology-node selection, and the literature gap addressed by this work.

Chapter 3 describes the proposed bidirectional RF front-end design. The overall architecture, asymmetric first gain stage, symmetric bidirectional gain stages, current-steering operation, neutralization technique, transformer-based matching circuits, and electromagnetic layout implementation are discussed.

Chapter 4 presents the simulation results and discussion. Receiver-mode and transmitter-mode S-parameters, noise figure, gain, output compression, reverse isolation, stability, power consumption, performance summary, and state-of-the-art comparison are included.

Chapter 5 concludes the thesis and discusses possible future extensions, including multi-channel beamformer integration, antenna-in-package implementation, top-level electromagnetic verification, and measurement validation.

2. SYSTEM-LEVEL DESIGN CONSIDERATIONS FOR 5G FR2 BEAMFORMING ARRAYS

2.1 RF Front-End Performance Metrics

The performance of a millimeter-wave front-end channel is evaluated using several metrics that directly affect the link budget and array-level operation. In the receiver mode, small-signal gain, input/output matching, reverse isolation, noise figure (NF), and linearity determine the sensitivity and dynamic range of the channel. In the transmitter mode, gain, output 1-dB compression point (OP1dB), saturated output power (Psat), efficiency, matching, and isolation determine the radiated power that can be obtained from an array.

For the proposed bidirectional front-end, these metrics must be interpreted at the channel level rather than only at the standalone amplifier level. Increasing the output power of a single RF channel generally requires larger device dimensions and higher bias currents. These requirements increase both silicon area and power dissipation, resulting in a higher thermal density within a limited chip area. Consequently, phased-array front-end channels should be evaluated not only in terms of achievable output power but also with respect to area efficiency, thermal constraints, power consumption, and array-level scalability.

2.2 Millimeter-Wave 5G Systems and Challenges

2.2.1 FR2 frequency bands

The 24–30 GHz frequency range overlaps with important 5G FR2 bands such as n257, n258, and n261. These bands are used for high-data-rate mobile access, fixed wireless access, small-cell deployments, and phased-array communication links. The wide available bandwidth in this frequency range enables high data rates, but the increased propagation loss necessitates antenna arrays with beamforming capability.

Table 2.1 summarizes the representative FR2 bands covered by the target design frequency range.

Table 2.1: Representative 5G FR2 bands covered by the target frequency range.

Band	Frequency Range	Relevance to This Work
n257	26.5–29.5 GHz	Covered
n258	24.25–27.5 GHz	Covered
n261	27.5–28.35 GHz	Covered

The selected 24–30 GHz frequency range therefore covers the lower and upper parts of the 28-GHz FR2 allocation. This makes the proposed front-end suitable for multi-band 5G phased-array applications where a single RF front-end channel should cover more than one operating band.

2.2.2 Path loss and the need for beamforming

The use of millimeter-wave bands in 5G systems is mainly motivated by the availability of wide contiguous bandwidth. However, compared to sub-6 GHz systems, FR2 links experience significantly larger free-space path loss and are more sensitive to blockage, antenna misalignment, and packaging losses. The free-space received power can be estimated using Friis' transmission equation:

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2, \tag{2.1}$$

where P_t and P_r are the transmitted and received powers, G_t and G_r are the transmitter and receiver antenna gains, λ is the wavelength, and R is the link distance. Since λ becomes small at FR2 frequencies, the path loss is high for a given physical link distance. Beamforming is therefore required to recover link margin through antenna directivity and coherent combining.

The implication for RFIC design is that the front-end cannot be considered independently from the antenna array. The output power, noise figure, gain, area, and power consumption of a single front-end channel must be evaluated together with the number of antenna elements that will be used in the final array.

2.2.3 Phased-array operating principles

A phased array consists of multiple antenna elements whose excitation phases and amplitudes are controlled to form a directive radiation pattern. If the spacing between adjacent elements is selected close to $\lambda/2$, the array can steer the main beam over a useful angular range while avoiding grating lobes for typical scan angles. A simplified linear phased-array concept is shown in Figure 2.1.

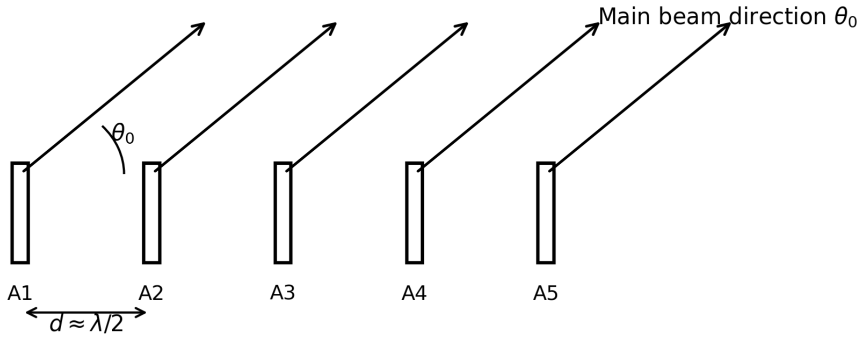

Figure 2.1: Simplified phased-array beam steering principle. The relative phase between adjacent antenna elements controls the main beam direction.

For a uniform linear array with element spacing d, steering the beam toward an angle θ_0 requires a progressive phase shift between adjacent elements. A commonly used expression for the required phase progression is

$$\Delta\phi = -k_0 d \sin(\theta_0), \tag{2.2}$$

where $k_0 = 2\pi/\lambda$ is the free-space wavenumber. This expression shows that beam steering is achieved by controlling the phase of each RF channel rather than mechanically rotating the antenna. In practical phased-array ICs, this phase control can be implemented at RF, LO, IF, or baseband depending on the selected architecture.

2.2.4 Beamforming architectures

Integrated 5G phased arrays can be implemented using analog, digital, or hybrid beamforming architectures. A conceptual comparison is shown in Figure 2.2. Each approach provides a different trade-off between flexibility, power consumption, area, and hardware complexity.

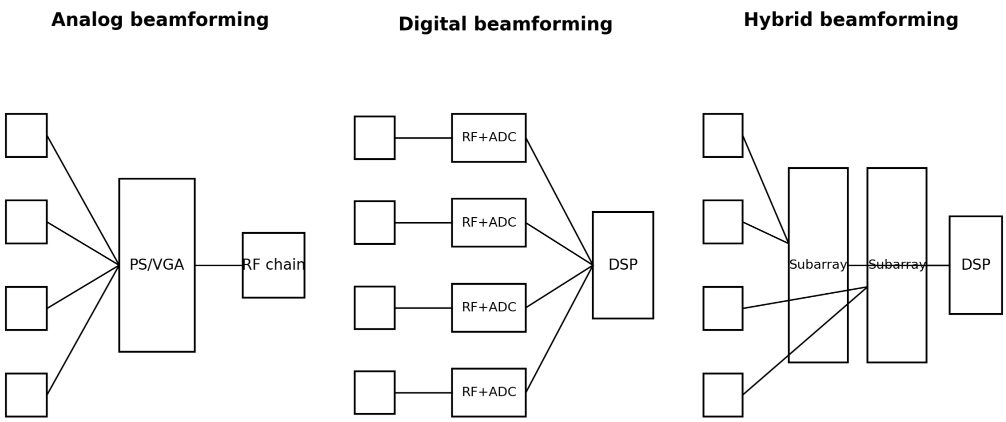

Figure 2.2: Conceptual comparison of analog, digital, and hybrid beamforming architectures.

In analog beamforming, multiple antenna elements share a single RF chain, and the beam is formed by RF phase shifters and variable-gain elements. This approach is attractive for compact mm-wave front-ends because it minimizes the number of mixers, local oscillator buffers, data converters, and baseband paths. In digital beamforming, each antenna element has its own complete RF chain and data converter, which provides the highest spatial flexibility but substantially increases area and power consumption at FR2 frequencies. Hybrid beamforming combines these two approaches by using several analog subarrays connected to multiple digital streams.

Table 2.2 summarizes the qualitative trade-off between these beamforming approaches for FR2 implementations.

Table 2.2: Qualitative comparison of beamforming architectures for FR2 systems.

Architecture	Main Advantage	Main Limitation
Analog beamforming	Low RF-chain count; compact and power efficient	Limited simultaneous beams and spatial flexibility
Digital beamforming	Maximum flexibility and multi-user capability	High power, area, and data-converter complexity
Hybrid beamforming	Balanced trade-off between flexibility and implementation cost	Requires both RF and baseband beam management

The proposed front-end is most naturally interpreted as a channel-level RF building block for analog or hybrid beamforming arrays. Since the circuit is bidirectional and intended for TDD operation, the same front-end channel can be reused in TX and RX modes, reducing duplicated passive and active circuitry.

2.2.5 Beamformer channel concept

In a practical phased-array IC, a beamformer channel typically includes the antenna interface, T/R front-end, phase control, and a combining or distribution network. Depending on the selected architecture, the phase-shifting function may be placed before or after the front-end block. Figure 2.3 illustrates a simplified beamformer channel representation and identifies the role of the proposed bidirectional front-end.

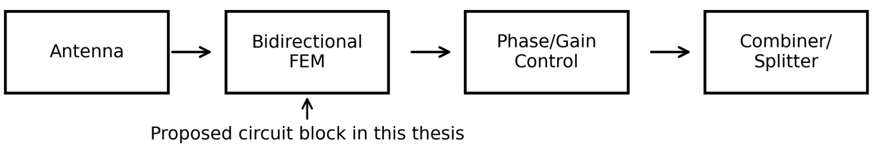

Figure 2.3: Simplified beamformer channel concept. The proposed bidirectional FEM can be used as the antenna-side RF front-end channel in a phased-array transceiver.

In this thesis, the proposed 24–30 GHz bidirectional front-end is not evaluated only as an isolated amplifier. Instead, it is considered as a compact antenna-side channel that can be replicated in a multi-channel beamformer IC. From this perspective, the active area, bidirectional operation, transformer-based matching circuits and moderate output power are all system-relevant design choices.

2.3 System-Level Requirements for Beamformer Channels

2.3.1 EIRP requirements and use cases

Effective isotropic radiated power (EIRP) is the main system-level metric that links a front-end channel to the radiated power of the full array. For a coherently combined array, the array-level EIRP can be approximated as

$$\mathrm{EIRP}_{\mathrm{dBm}} = P_{out,ch,\mathrm{dBm}} + G_{elem,\mathrm{dBi}} + 20\log_{10}(N), \tag{2.3}$$

where $P_{out,ch}$ is the output power per channel, G_{elem} is the gain of each antenna element, and N is the number of coherently combined antenna elements. This expression shows that increasing the number of array elements relaxes the required output power of each individual front-end channel.

The required EIRP strongly depends on the deployment scenario. Table 2.3 gives a qualitative interpretation of typical FR2 use cases and the corresponding front-end design implication.

Table 2.3: Qualitative interpretation of EIRP-driven FR2 use cases.

Use Case	Typical Design Priority	Implication for Front-End Channel
User equipment	Low power and small form factor	Low-to-moderate channel power with a compact array
CPE/FWA terminal	Higher aperture and improved link margin	Moderate channel power with a larger array
Small cell	Outdoor coverage and multi-user operation	Moderate-to-high EIRP through more elements or higher channel power
Macro base station	Long-range coverage and high sector power	High array count and/or higher per-channel output power

Therefore, the same front-end channel can correspond to different system classes depending on how many times it is replicated in the antenna array. Increasing the per-channel output power would generally require larger output devices and higher bias currents, which would increase silicon area, DC power consumption, and local thermal density. For this reason, the output-power target must be evaluated together with area efficiency, heat dissipation, and array-level scalability rather than as an isolated single-channel metric.

Figure 2.4 illustrates the required per-element output power as a function of the number of coherently combined array elements for representative EIRP targets. The figure is redrawn and adapted from the system-level array-scaling analysis in [6]. The shaded region highlights the approximate output-power range around the proposed FEM capability.

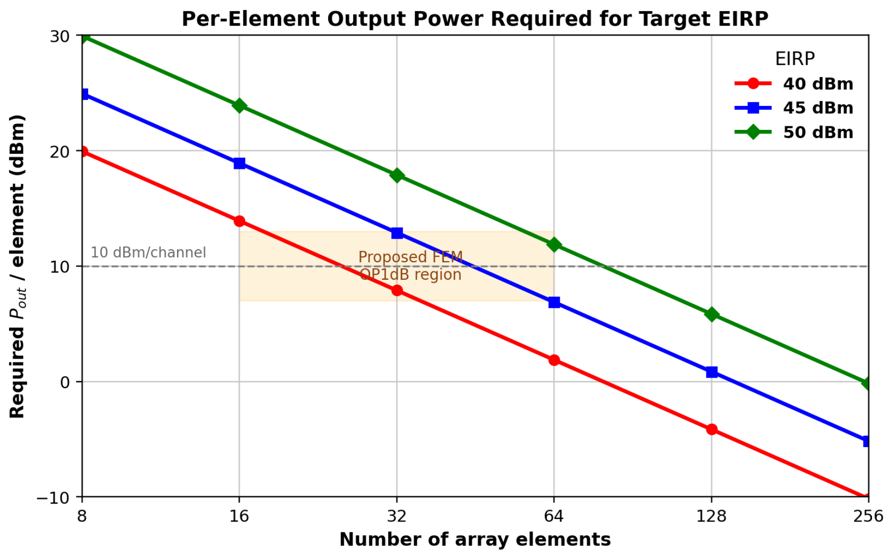

Figure 2.4: Required per-element output power for representative EIRP targets as a function of array size, adapted from the array-scaling analysis in [6].

The figure clarifies how the required per-channel output power scales with array size. For a given EIRP target, coherent combining relaxes the power required from each individual front-end channel. This relation is important for compact CMOS implementations because excessive per-channel output power increases device size, current consumption, routing complexity, and local thermal density.

2.3.2 Output power requirements

The transmitter output power requirement is derived from the EIRP target, antenna element gain, and number of coherently combined elements. For example, assuming a channel OP1dB of 10 dBm and an antenna element gain of 5 dBi, the estimated array EIRP values for different array sizes are summarized in Table 2.4.

Table 2.4: Estimated array EIRP using a 10 dBm front-end channel and 5 dBi antenna element gain.

Number of Elements	Coherent Term	Element Gain	Estimated EIRP
16	24.1 dB	5 dBi	39.1 dBm
32	30.1 dB	5 dBi	45.1 dBm
64	36.1 dB	5 dBi	51.1 dBm
128	42.1 dB	5 dBi	57.1 dBm

These estimates show that the proposed front-end output-power level is more meaningful when interpreted as one element of a coherent array. The 16-element case is close to compact terminal or short-range links, while 32- and 64-element cases become more relevant for CPE/FWA and small-cell-class links. Macro base-station-class operation would typically require either a larger array, higher element gain, additional output-stage power, or a combination of these.

2.3.3 Receiver noise figure requirements

The receiver noise figure is the second critical system-level metric after output power. While EIRP determines the transmit-side link margin, receiver NF determines how much received signal power is required to achieve a given signal-to-noise ratio. Beamforming increases the received signal through array gain, but it does not remove the need for a low-noise front-end because the first active stages still dominate the receiver noise contribution.

Reported Ka-band CMOS LNAs, beamformer channels, and bidirectional front-end modules commonly achieve noise figures within an approximate 3–7 dB range, depending on bandwidth, gain, matching loss, technology node, and integration level

[3]–[5]. Instead of comparing only individual works, Table 2.5 gives a qualitative interpretation of this practical design range.

Table 2.5: Qualitative interpretation of Ka-band CMOS receiver noise figure regions.

Noise Figure Region	Interpretation
<4 dB	Aggressive target for highly optimized low-noise channels
4–6 dB	Typical region for Ka-band CMOS beamformer and FEM channels
6–8 dB	Acceptable but with reduced receiver sensitivity margin
>8 dB	Generally undesirable for sensitive FR2 receivers

The proposed front-end achieves a simulated noise figure of approximately 4.4–5.5 dB across the operating band. Therefore, it falls within the typical Ka-band CMOS receiver region while simultaneously supporting bidirectional operation, compact layout, and wideband 24–30 GHz coverage. The design does not target the absolute minimum achievable NF; instead, it balances receiver noise performance with TX/RX hardware reuse and compact front-end integration.

2.4 Bidirectional Front-End Architectures

Conventional millimeter-wave TDD front-ends generally use separate PA and LNA paths that are connected to a shared antenna through a T/R switch. This approach allows the PA and LNA to be optimized independently, but it also duplicates matching circuits and introduces switch insertion loss. At FR2 frequencies, the switch loss directly degrades the TX output power and the RX noise figure, while the additional passive networks increase the area of each array element. Transformer-based T/R-switch front-ends reduce this penalty by combining the switch, balun, and matching functions within a coupled passive structure. For example, a 24.25–27.5 GHz 65-nm CMOS FEM uses a transformer-based TRSW that also realizes balun and matching functions, achieving 14.4 dBm OP1dB and 16 dBm Psat in TX mode while maintaining 5.9 dB minimum NF in RX mode [8]. A separate PA/LNA/SPDT solution can also be optimized for high linearity and load-variation robustness, as demonstrated in a 24–30 GHz SiGe TRX front-end with an asymmetric SPDT switch and balanced PA [14].

An alternative approach is to remove the conventional standalone switch and share a bidirectional signal path between TX and RX modes. In this class, the PA/LNA cores and matching circuits are co-designed so that the same passive interface can be used in both directions. A 90-nm CMOS switchless PA-LNA uses a current-type transformer as a bidirectional matching circuit and avoids the loss of a separate T/R switch [9]. A broadband hybrid N/PMOS bidirectional PA/LNA further extends this concept by using different device types and shared matching to achieve wideband operation from 26 to 39 GHz [5]. More recent compact bidirectional PA-LNAs employ stacked PA structures and shared transformer-based matching circuits to increase output power while maintaining compact area and acceptable receiver noise performance [16].

The two architecture families therefore represent different design trade-offs. The T/R-switch-based front-end preserves a more conventional separation between PA and LNA paths, but it requires a switch and additional passive interfaces. The shared-path bidirectional front-end reduces duplicated hardware and can be more area efficient, but it requires joint optimization of PA load impedance, LNA source impedance, isolation, stability, and biasing. The proposed architecture follows the shared-path design philosophy. Unlike single-core PA-LNA examples, however, it uses multiple transformer-coupled bidirectional gain stages so that the complete RF front-end path can be reused across TX and RX modes.

2.5 Transformer-Based Matching Circuits

Transformer-based matching circuits are used in this thesis as compact passive interfaces between the antenna port, the bidirectional gain stages, and the shared RF signal path. Their role is not limited to impedance transformation. In differential CMOS front-ends, an on-chip transformer can also provide differential signal interfacing, dc isolation between cascaded stages, center-tap biasing, and balun functionality. Bloch and Socher emphasize that these properties make transformers a natural choice for cascaded CMOS RF and millimeter-wave blocks [12].

In bidirectional front-ends, transformer-based passives have evolved from simple matching elements into multifunctional building blocks. In transformer-assisted TRSW front-ends, the transformer can be reused as a switch, balun, PA output matching circuit, and LNA input matching circuit within a compact footprint [8]. In switchless bidirectional PA-LNAs, current-type transformers can serve as the shared bidirectional matching interface between the PA and LNA ports [9]. Ultra-compact Ka-band FEMs further exploit folded or magnetically self-canceling transformer layouts to stack or fold TX and RX passive paths into the same area, reducing the core size to the 0.06–0.1 mm^2 range [4,15]. These examples show that transformer-based matching is not only a circuit-level matching technique, but also an architectural mechanism for passive reuse and area reduction.

Traditional microwave matching can be implemented using transmission-line sections, lumped LC networks, or transformer-based structures. At Ka-band frequencies, transmission-line sections may occupy a large silicon area because the required electrical lengths become a significant fraction of the on-chip wavelength. Lumped LC networks can be more compact, but additional passive structures may still be needed when impedance transformation, differential interfacing, and biasing must be achieved simultaneously. Transformer-based matching provides a compact alternative because magnetic coupling introduces additional design variables, mainly the coupling coefficient and the winding ratio, while supporting the differential circuit implementation required in the proposed front-end.

2.5.1 Transformer Matching Theory

The transformer synthesis method used in this thesis is based on the first-order coupled-inductor model shown in Figure 2.5. In this model, L_1 and L_2 denote the primary and secondary winding inductances, while k denotes the magnetic coupling coefficient.

The physical range of the coupling coefficient is

$$0 < k < 1. \tag{2.4}$$

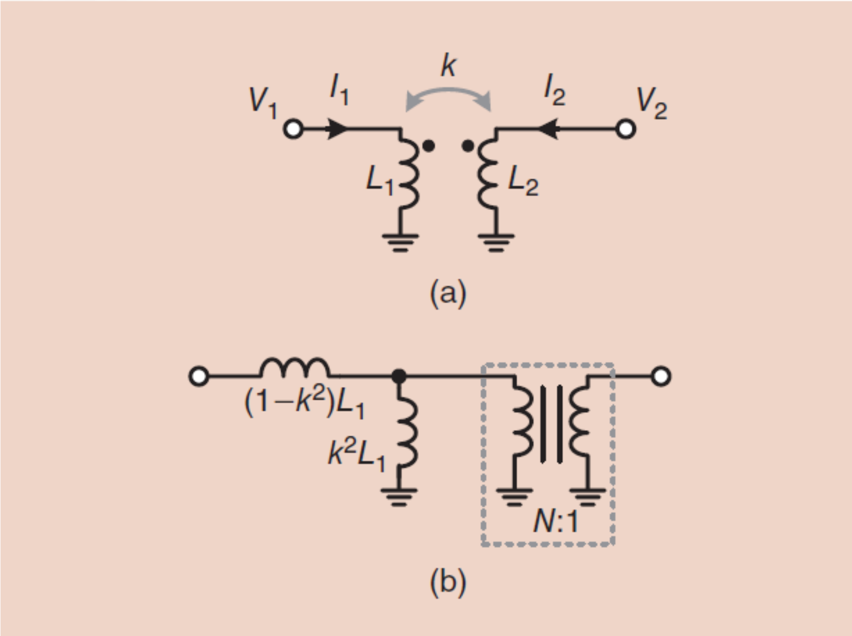

Figure 2.5: First-order transformer model and equivalent circuit representation adapted from [12].

For practical millimeter-wave CMOS transformers, values around $k = 0.6$–0.8 are commonly used as an initial design range. Lower values indicate weak magnetic coupling and larger leakage effects, whereas values close to unity represent a more ideal transformer but may require a tighter layout and stronger magnetic coupling. Following the notation of [12], the leakage-related parameter is defined as

$$\alpha = \frac{1-k^2}{k^2}. \tag{2.5}$$

The parameter α represents the effect of imperfect magnetic coupling in the equivalent circuit. As k approaches unity, α approaches zero and the transformer behaves more ideally. Conversely, weaker coupling increases α, which means that the leakage-inductance contribution becomes more significant. The auxiliary inductance used in the equivalent model is defined as

$$L = k^2 L_1. \tag{2.6}$$

The matching formulation assumes that both the source and load impedances are capacitive. This assumption is suitable for many CMOS RF stages because MOS gates, drains, interconnects, and pad interfaces often present capacitive reactance around the design frequency. Therefore, the source and load impedances are written as

$$Z_S = R_S - jQ_S R_S, \tag{2.7}$$

$$Z_L = R_L - jQ_L R_L, \tag{2.8}$$

where R_S and R_L are the real parts of the source and load impedances, and Q_S and Q_L are the corresponding positive quality factors. The formulation directly covers capacitive-to-capacitive and capacitive-to-real matching cases, but it does not provide a direct solution when both impedances are inductive.

Figure 2.6 defines the transformer matching problem used in the analytical formulation. The objective is to choose the transformer parameters such that the impedance looking into the matching circuit becomes the conjugate of the source impedance at the target frequency.

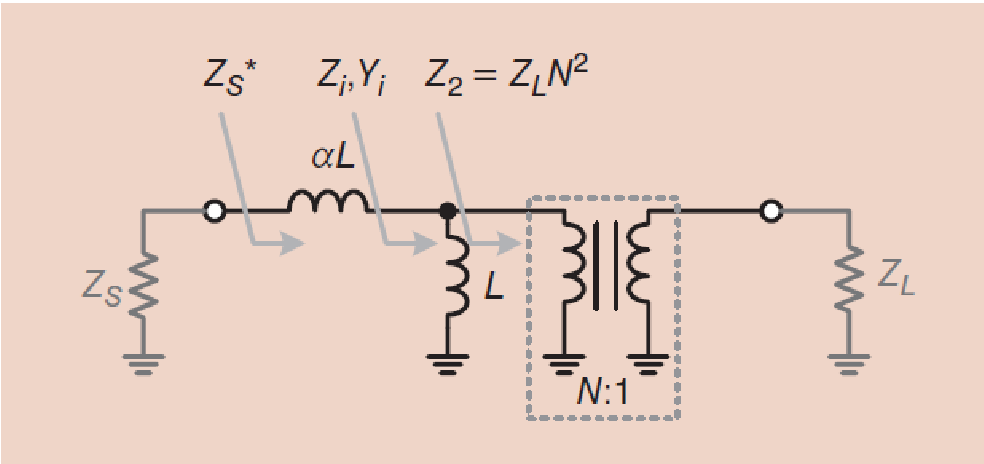

Figure 2.6: Equivalent transformer matching representation with source, load, and intermediate impedance definitions adapted from [12].

The variables appearing in Figure 2.6 are summarized in Table 2.6. This table is included to make the notation used in the analytical derivation explicit before the design procedure is introduced.

Using the equivalent representation, Bloch and Socher show that the transformer matching problem can be reduced to normalized winding quality factors. Instead of directly solving for L_1 and L_2, the method first obtains

$$Q_{XL1} = \frac{\omega L_1}{\mathrm{Re}\{Z_S\}}, \tag{2.9}$$

Table 2.6: Symbols used in the transformer matching formulation.

Symbol	Description
Z_S	Source impedance to be conjugately matched
Z_L	Load impedance connected to the transformer secondary side
Z_i	Intermediate impedance used in the matching condition
Y_i	Intermediate admittance corresponding to Z_i
N	Ideal transformer turns ratio in the equivalent model
k	Magnetic coupling coefficient between transformer windings
α	Leakage-related parameter caused by nonideal coupling
L_1, L_2	Primary and secondary winding inductances
Q_S, Q_L	Source and load impedance quality factors

$$Q_{XL2} = \frac{\omega L_2}{\mathrm{Re}\{Z_L\}}. \tag{2.10}$$

These cross-quality factors provide normalized transformer targets that are later converted into physical inductance values. The detailed derivation of the normalized impedance variable, the matching equation, and the closed-form expressions used to obtain Q_{XL1} and Q_{XL2} is given in Appendix A. In the main text, only the resulting design flow is used, because the purpose of this section is to explain how the method supports the proposed RF front-end design rather than to reproduce the complete mathematical proof.

2.5.2 Transformer Matching Design Procedure

The transformer-based matching approach can be used as an initial synthesis procedure before full electromagnetic optimization. The practical design flow is summarized in Table 2.7. The result of this procedure is not the final layout, but a physically meaningful starting point for EM-based transformer tuning.

After the normalized winding quality factors are obtained, the corresponding inductance values are calculated as

$$L_1 = \frac{Q_{XL1} R_S}{\omega}, \tag{2.11}$$

$$L_2 = \frac{Q_{XL2} R_L}{\omega}. \tag{2.12}$$

These values are then used as the starting point for the physical transformer geometry. In the final implementation, the winding width, spacing, metal layer selection,

Table 2.7: Transformer matching design procedure used as an initial synthesis flow.

Step	Description
1	Extract the source impedance Z_S at the target frequency.
2	Extract the load impedance Z_L at the target frequency.
3	Compute Q_S and Q_L from the real and imaginary parts of the impedances.
4	Select an initial coupling coefficient, typically $k = 0.6$–0.8 for mm-wave CMOS.
5	Determine Q_{XL1} and Q_{XL2} using the normalized matching formulation.
6	Convert the normalized values into L_1 and L_2 using ω, R_S, and R_L.
7	Implement the transformer layout and optimize k, Q, SRF, insertion loss, and matching by EM simulation.

underpass routing, symmetry, self-resonance frequency, and substrate loss must be verified through electromagnetic simulation. Therefore, the analytical method provides the initial transformer target, while EM optimization determines the final layout values used in the proposed 24–30 GHz bidirectional front-end.

2.6 Neutralization Techniques

At millimeter-wave frequencies, the gate-drain capacitance C_{gd} of a MOS transistor introduces reverse coupling from the drain to the gate. This feedback degrades gain, worsens reverse isolation, and can reduce stability margin. A simple way to understand capacitive neutralization is to examine the small-signal current through C_{gd}. For a common-source device, the feedback current associated with the intrinsic gate-drain capacitance can be written as

$$i_{gd} = j\omega C_{gd}(v_g - v_d). \tag{2.13}$$

In a differential amplifier, an intentional cross-coupled neutralization capacitance C_N can be inserted between the drain of one side and the gate of the opposite side. Under differential excitation, the neutralization current has the opposite sign with respect to the intrinsic C_{gd} feedback current. The net feedback current can therefore be expressed as

$$i_{fb} = j\omega(C_{gd} - C_N)(v_g - v_d). \tag{2.14}$$

This expression shows that the neutralization capacitor acts as an effective reduction of the gate-drain capacitance,

$$C_{gd,\text{eff}} = C_{gd} - C_N. \tag{2.15}$$

In the ideal case, choosing

$$C_N = C_{gd} \tag{2.16}$$

cancels the first-order capacitive feedback, resulting in $C_{gd,\text{eff}} = 0$. In practice, C_N is tuned below or around the effective parasitic value because over-neutralization can create positive feedback and reduce stability. Therefore, the neutralization capacitor must be co-designed with the device size, layout parasitics, matching circuits, and bias condition.

Bidirectional front-ends use this principle in different ways. Conventional differential PA-LNAs may employ explicit cross-coupled capacitors [9]. Hybrid N/PMOS bidirectional cores use the off-state device parasitic capacitance as a neutralization element in the opposite mode [5]. Neutralized bidirectional beamformer chains extend the same idea to multiple cascaded stages for shared TX/RX operation [3]. Stacked PA-LNA structures also exploit parasitic capacitances and device stacking to improve PA output capability, LNA stability, and matching compatibility [16]. In the proposed design, neutralization is used as a compact stability and reverse-isolation enhancement method for the shared bidirectional gain stages.

2.7 Literature Comparison and Research Gap

The literature shows a clear evolution from conventional T/R-switch-based front-ends toward increasingly compact shared-path bidirectional implementations. Transformer-based TRSW designs reduce the penalty of the antenna switch by merging switching, balun, and matching functions [8]. Switchless bidirectional PA-LNAs further remove the standalone T/R switch and share the PA/LNA passive interface [5,9]. Ultra-compact FEMs then use folded or stacked transformer layouts to further reduce the passive footprint [4,15]. Finally, neutralized and stacked bidirectional

PA-LNAs improve gain, linearity, stability, and output power by exploiting device parasitics and bias reconfiguration [3,16].

Table 2.8 summarizes representative mm-wave bidirectional front-ends and related TRX FEMs. The comparison highlights that prior works typically emphasize one dominant advantage, such as high output power, low NF, broad bandwidth, or ultra-compact area. The objective of this thesis is different: it investigates the feasibility of a 24–30 GHz, 65-nm CMOS, transformer-coupled bidirectional front-end architecture that reuses multiple active gain stages and matching circuits while maintaining balanced TX and RX performance.

Table 2.8: Comparison with representative mm-wave bidirectional front-ends and TRX FEMs.

Work	Topology	Tech.	Freq.	RX Gain / NF	TX Gain	OP1dB / Psat	Area
[14]	SPDT TRX FEM	0.13-um SiGe	24–30	26 / <5.3	–	>15.9 / –	–
[8]	TF-TRSW FEM	65-nm CMOS	24.25–27.5	25.5 / 5.9	35	14.4 / 16	0.26
[9]	Switchless PA-LNA	90-nm CMOS	30–39.2	17.6 / 4.7	18.3	13.3 / 15.1	0.21
[5]	Hybrid N/PMOS PA-LNA	45-nm SOI	26–39	17.6 / 5.2	18.9	>16.3 / 19.4	0.19
[15]	Folded TF FEM	65-nm CMOS	Ka-band	26 / 4.0	26.5	13.5 / 15.1	0.10
[4]	T/R-stacked FEM	65-nm CMOS	Ka-band	18.2 / 3.8	26	13.5 / 15.1	0.06
[16]	Stacked PA-LNA	28-nm CMOS	27.3–35.4	17.3 / 5.3	20.4	15.0 / 17.4	0.10
This work	Shared-path FEM	65-nm CMOS	24–30	19–24 / 4.4–5.5	TBD	TBD	≈0.18

Notes: Gain and NF are in dB, OP1dB and Psat are in dBm, and area is in mm^2. Values are reported or estimated from the cited works; TBD entries will be updated after final TX-mode result extraction.

Compared with T/R-switch-based front-ends, the proposed architecture aims to reduce duplicated TX/RX signal paths. Compared with single-core bidirectional PA-LNAs, it extends the shared-path idea to a multi-stage front-end module with transformer-coupled interstage interfaces. Compared with ultra-compact folded-transformer FEMs, this work emphasizes the feasibility of a three-stage bidirectional architecture in a standard 65-nm CMOS process with transformer-based matching and capacitive neutralization. This gap motivates the proposed architecture in Chapter 3.

2.8 Technology Node Selection and Cost Considerations

Technology-node selection in a mm-wave front-end is not determined solely by transistor speed. Advanced CMOS nodes provide higher f_T and f_{max}, but they also introduce higher fabrication cost, lower voltage headroom, and increased design complexity. In addition, the area of mm-wave RFICs is often dominated by passive structures such as transformers, baluns, transmission lines, pads, and ESD networks. Therefore, moving from 65 nm to a more advanced node does not necessarily reduce the total front-end area in proportion to the lithographic scaling factor.

The technology-node trade-off is summarized qualitatively in Table 2.9. The cost trend follows the general observation that advanced nodes such as 28 nm have higher wafer, mask, and design costs than mature nodes such as 65 nm, while mm-wave RF performance benefits from higher transistor speed [10,11,13].

Table 2.9: Qualitative comparison of CMOS technology nodes for mm-wave front-end implementations.

Node	Relative Cost	RF Performance	Passive Scaling
130 nm	Very low	Limited for compact FR2 channels	Poor
90 nm	Low	Moderate	Poor-to-moderate
65 nm	Moderate	Sufficient for 24–30 GHz RFICs	Moderate
45/40 nm	High	Higher transistor speed	Limited by passives
28 nm	Very high	Excellent transistor speed	Limited by passives and voltage headroom

Table 2.9 indicates that 65 nm CMOS provides a balanced choice for the proposed front-end. It offers sufficient RF performance for 24–30 GHz operation, mature RF modeling, thick top-metal options for transformer implementation, and lower cost than more advanced technology nodes. Since the proposed design is dominated by transformer-based matching and compact layout co-design rather than by digital logic density, 65 nm CMOS is a reasonable technology choice for this thesis.

2.9 Chapter Summary

This chapter established the system and technology background required to understand the proposed 24–30 GHz bidirectional front-end. The FR2 frequency bands motivate the use of phased arrays, while the beamformer channel concept explains how the proposed FEM can be interpreted as one repeatable RF channel in a larger array. The EIRP and output-power analysis show that a 10 dBm-class channel can be meaningful when replicated across a coherent array, particularly for compact arrays, CPE/FWA, and small-cell-class systems. The NF discussion shows that the simulated 4.4–5.5 dB receiver NF lies within the typical Ka-band CMOS front-end region. The literature review then compared T/R-switch-based and shared-path bidirectional front-end topologies, transformer-based passive reuse, and capacitive neutralization. Finally, the technology-node discussion motivates the use of 65 nm CMOS for the proposed transformer-coupled bidirectional architecture described in the following chapter.

3. PROPOSED BIDIRECTIONAL RF FRONT-END DESIGN

3.1 Overall Architecture

The proposed RF front-end is designed as a compact bidirectional module operating over the 24–30 GHz 5G FR2 frequency range. The architecture consists of three cascaded active gain stages and transformer-based matching circuits. The signal path is reused in transmitter and receiver modes by changing the bias conditions and steering the operating direction of the active stages.

Figure 3.1 shows the top-level architecture of the proposed three-stage transformer-based bidirectional front-end. The antenna side and the common-path side are connected through a fully shared differential signal chain. The same active stages and passive matching circuits are therefore used in both PA and LNA modes, while the circuit operates in a half-duplex manner. This organization is intended to reduce the area penalty that would otherwise result from using separate PA, LNA, and matching circuits in each beamformer channel.

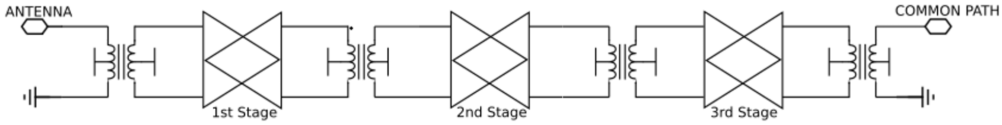

Figure 3.1: Proposed three-stage transformer-based bidirectional front-end architecture.

Placeholder: This subsection will be expanded with a more detailed explanation of the complete signal flow in PA and LNA modes, the role of each transformer-based matching circuit, and the relation between this single-channel front-end and a multi-channel beamformer IC.

3.2 First Gain Stage Design

The first gain stage is the most critical block because it directly affects both receiver noise performance and transmitter output capability. In the proposed architecture, this stage is intentionally asymmetric. In receiver mode, the LNA-side device is optimized for low noise and input matching. In transmitter mode, the PA-side device is optimized for voltage swing, output power, and load driving capability.

Figure 3.2 illustrates the asymmetric bidirectional first-stage topology. The PA path uses a cascode configuration on the antenna side to improve voltage handling capability, whereas the LNA path uses a simpler gain cell to avoid unnecessary parasitic loading in receive operation. The signal direction is controlled by tail-current switching rather than supply switching. Therefore, the core gate bias voltages can remain fixed, which helps preserve safe operating area conditions.

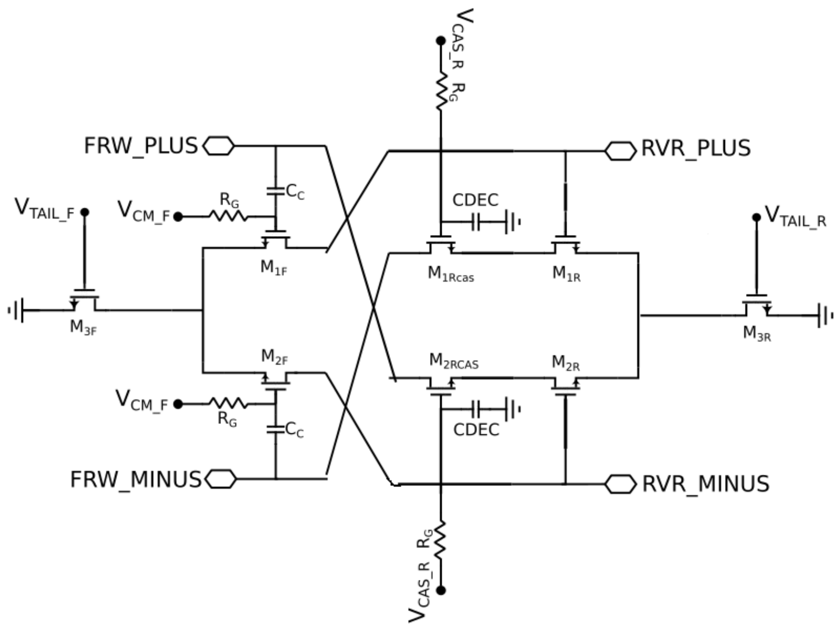

Figure 3.2: Asymmetric bidirectional first-stage topology with capacitive neutralization.

3.2.1 Receiver Mode Operation

In receiver mode, the input signal is received from the antenna-side matching circuit and amplified through the low-noise path. Since the first stage dominates the cascaded receiver noise figure, the transistor sizing, bias current, and source impedance are selected by considering both noise matching and input return loss. The inactive PA devices remain biased but do not conduct the main signal current.

Placeholder: Add the detailed receive-mode small-signal path, noise matching discussion, and explanation of how the OFF-state PA parasitics contribute to LNA-mode neutralization.

3.2.2 Transmitter Mode Operation

In transmitter mode, the signal propagates in the reverse direction and the antenna-side cascode device operates as the output-driving device. The cascode configuration increases voltage handling capability and supports a larger RF swing at the output. This is especially important because the first stage is directly connected to the antenna-side matching circuit and therefore has a strong effect on the transmitter output compression point.

Placeholder: Add the PA-mode load-line discussion, voltage division across the cascode devices, and the sizing procedure for the external gate capacitor used for safe voltage distribution.

3.2.3 Asymmetric Architecture

The asymmetric first stage is introduced because the optimum transistor dimensions and impedance environments for PA and LNA operation are different. A PA favors larger voltage swing and higher output capability, while an LNA favors low noise contribution and reduced parasitic loading. The proposed architecture therefore avoids forcing the same device structure to satisfy both requirements equally.

Placeholder: Add comparison with a fully symmetric bidirectional stage and explain why asymmetry improves the PA–LNA trade-off in the first stage.

3.3 Bidirectional Gain Stage Design

The second and third gain stages are implemented as shared bidirectional stages. These stages provide additional gain in both operating directions and distribute the total gain requirement across the complete front-end. Unlike the first stage, these stages use a symmetric cross-coupled structure because they are not directly constrained by the antenna-side noise and output-power trade-off.

Figure 3.3 shows the symmetric cross-coupled bidirectional gain cell used in the second and third stages. The signal direction is selected by activating the corresponding tail-current device. This avoids supply switching and enables compact replication of the gain cell in a multi-stage front-end.

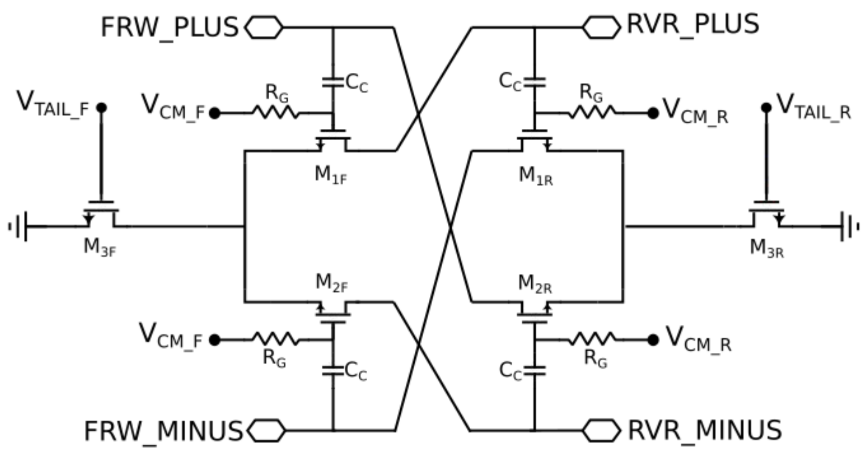

Figure 3.3: Second and third stages implemented as symmetric cross-coupled bidirectional gain cells.

3.3.1 Current-Steering Mechanism

The operating direction is controlled by biasing the corresponding tail-current devices and steering current through the desired path. This approach preserves a shared signal path while keeping the inactive devices in a controlled bias condition.

Placeholder: Add a mode table showing the ON/OFF state of each tail device for PA and LNA operation.

3.3.2 Shared Signal Path

The shared signal path reduces area by reusing passive matching structures and active devices. However, this also requires careful isolation and stability analysis because inactive devices can introduce parasitic loading and feedback paths.

Placeholder: Add discussion on how shared interstage transformers are optimized for bidirectional propagation.

3.3.3 Driver Requirement for PA Operation

The second and third stages must provide sufficient drive capability for the output PA stage. In particular, their available output power should exceed the PA input 1-dB compression level so that the final stage can reach its intended output compression point.

Placeholder: Add the simulated interstage drive levels and compare them with the PA IP1dB requirement.

3.4 Mode-Control and Biasing Strategy

The operating mode of the proposed front-end is selected by steering the bias current through the desired signal path. In receive operation, the antenna-side low-noise path is activated and the reverse transmitter path is kept inactive. In transmit operation, the signal propagates in the opposite direction and the antenna-side output-driving devices are biased for larger RF swing.

This current-steering strategy avoids direct supply switching in the RF signal path. As a result, the gate and supply bias conditions can be kept well controlled in both operating modes, which is important for stability, repeatability, and safe device operation.

Placeholder: Add a bias-state table showing the active and inactive devices in RX and TX modes.

3.5 Neutralization Technique

Neutralization is used to reduce the effect of gate–drain capacitance and improve reverse isolation. At millimeter-wave frequencies, unwanted feedback through C_{gd} can degrade stability and limit the maximum achievable gain. The proposed design uses capacitive neutralization in both the asymmetric first stage and the symmetric bidirectional gain stages.

Figure 3.4 shows a simplified equivalent circuit of a neutralized differential gain cell. The cross-coupled capacitive path introduces an opposite-polarity feedback component that can partially cancel the intrinsic gate–drain feedback of the transistors. This improves unilateral behavior and increases stability margin.

Placeholder: Add the mathematical neutralization condition, the effect on G_{max}, and the stability-factor comparison with and without neutralization.

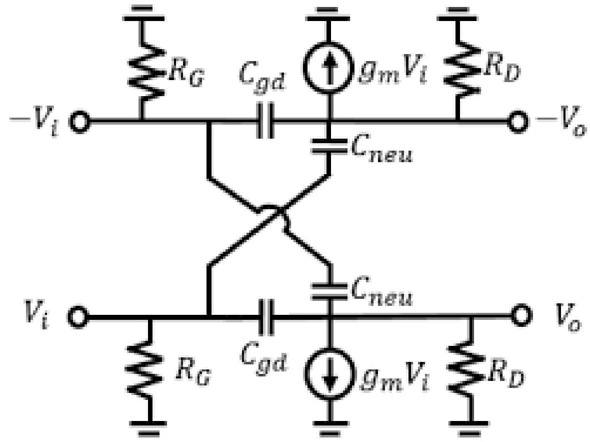

Figure 3.4: Simplified equivalent circuit illustrating capacitive neutralization in a differential gain cell.

3.6 Transformer-Based Matching Networks

Transformer-based matching circuits are used at the input, output, and interstage nodes. These structures provide impedance transformation, differential operation, DC biasing convenience, and compact area compared with distributed transmission-line implementations.

3.6.1 Input Matching Transformer

The input transformer is designed to provide receiver input matching while keeping the noise figure low. Its electromagnetic model should include metal loss, substrate coupling, parasitic capacitance, and coupling factor.

Placeholder: Insert the input transformer layout/EM view and summarize its extracted inductance, coupling factor, and insertion loss.

3.6.2 Interstage Matching Transformers

The interstage transformers tune the gain stages around the target band and provide differential coupling between stages. Their inductance values and coupling factors are selected to maintain bandwidth across 24–30 GHz.

Placeholder: Add the interstage transformer design table and explain how the PA and LNA impedance requirements are co-designed.

3.6.3 Output Matching Transformer

The output transformer is designed considering transmitter output power and receiver-mode loading. It must provide a suitable load for PA operation while not excessively degrading RX performance.

Placeholder: Add the output transformer EM result and discuss its role in PA load matching.

3.7 Electromagnetic Modeling and Layout Implementation

All transformer-based passive structures should be verified using electromagnetic simulations. The final layout should include device placement, RF routing, bias decoupling, ground shielding, and pad effects. The compact core area is a key metric because the proposed front-end is intended to be replicated in multi-channel beamformer ICs.

Placeholder: Add the top-level EM view, clarify which instances are included or excluded in the EM setup, and explain the expected difference between block-level EM and full top-level EM results.

4. SIMULATION RESULTS AND DISCUSSION

4.1 Receiver-Mode S-Parameter Results

Receiver-mode S-parameters are evaluated separately for the lower and upper LNA configurations. The most important metrics are forward gain, input matching, output matching, and reverse isolation. The final thesis version will include the selected plots for both frequency configurations.

4.1.1 Lower-Band LNA Configuration

For the 24.25–27.5 GHz lower-band configuration, the simulated forward gain is approximately 19–24 dB. The input matching reaches its best value around 26 GHz, while the output matching also shows its strongest response near the center of the band. The reverse isolation remains below approximately -105 dB in the present block-level simulation results.

The simulated response demonstrates adequate matching over the target FR2 band while maintaining high gain. The input reflection coefficient reaches its minimum near the center frequency, indicating successful co-design of the transformer-based matching circuit and the input transistor impedance. Similarly, the output return loss remains acceptable across the entire operating range. The reverse isolation is extremely high because the transmitter and receiver signal paths are separated and because only block-level electromagnetic simulations are considered at this stage.

4.1.2 Upper-Band LNA Configuration

For the 26.5–29.5 GHz upper-band configuration, the simulated gain is approximately 17.5–20 dB. The gain decreases gradually toward the upper edge of the band, while the input and output matching remain suitable for broadband FR2 operation.

The upper-band tuning state shifts the optimum matching point toward higher frequencies through switched passive elements. Although a small reduction in peak gain is observed, broadband coverage of the upper FR2 channels is achieved while preserving acceptable matching and noise performance.

4.2 Noise Figure Performance

The receiver noise performance is evaluated using both NF and NFmin. In the lower-band configuration, NFmin is approximately 4.2–4.8 dB and NF is approximately 4.4–5.5 dB. In the upper-band configuration, NFmin is approximately 4.4–5.0 dB and NF is approximately 4.7–5.3 dB.

The obtained NF and NFmin values are consistent with the transformer-based input matching strategy. The small increase in NF toward the band edges originates from the deviation between the optimum noise impedance and the impedance presented by the broadband matching circuit.

4.3 Receiver Linearity Performance

Receiver linearity should be evaluated using input compression and third-order intermodulation metrics. These results are important for blocker tolerance and receiver dynamic-range assessment.

Placeholder: Insert receiver IP1dB and IIP3 simulations after finalizing the large-signal receiver testbench.

4.4 Transmitter-Mode Performance

Transmitter-mode results characterize gain, output power, compression, and efficiency. The PA-mode response is evaluated over the full 24–30 GHz band.

4.4.1 Power Gain

The simulated transmitter gain reaches approximately 28 dB near the lower part of the operating range and gradually decreases toward 30 GHz. This behavior is consistent with the wideband but compact transformer-based matching strategy.

Placeholder: Insert the PA-mode S-parameter plot and discuss the matching and gain roll-off across the band.

4.4.2 Output Power Characteristics

The large-signal simulations indicate a peak saturated output power of approximately 13 dBm. The output power should be interpreted together with the compact core area and the bidirectional sharing of active and passive structures.

Placeholder: Insert large-signal output power and PAE curves at representative frequencies.

4.4.3 Output Compression Point

The simulated output 1-dB compression point reaches approximately 10 dBm around 25–26 GHz and decreases to about 8 dBm near 30 GHz. This value is important because it directly determines the array-level EIRP through coherent combining.

For a 16-channel array, a 10 dBm channel-level OP1dB corresponds to approximately 39 dBm EIRP when a 5 dBi element gain is assumed before modulation back-off.

Placeholder: Add the OP1dB/IP1dB plot and connect the result to the EIRP discussion in Chapter 2.

4.5 Reverse Isolation Performance

Reverse isolation is important for both stability and unwanted coupling suppression. The present simulations show very high reverse isolation; however, the final value should be interpreted carefully after top-level electromagnetic extraction because additional coupling paths may appear in the complete layout.

Placeholder: Add the final extracted reverse-isolation results and discuss the difference between schematic, block-level EM, and top-level EM simulations.

4.6 Stability Analysis

Stability must be verified over a broad frequency range, not only over the operating band. Rollett stability factor and μ factor simulations should be included for both PA and LNA modes.

Placeholder: Insert K-factor and μ-factor plots with and without neutralization.

4.7 Power Consumption Analysis

Power consumption is reported separately for receiver and transmitter modes. The current design consumes approximately 96 mW in RX mode and 196 mW in TX mode. RX-mode current is important for array-level power scaling, while TX-mode power consumption must be interpreted together with output power and PAE.

Placeholder: Add a bias table listing supply voltages, branch currents, and total power in each operating mode.

4.8 Performance Summary

Table 4.1 summarizes the current simulated performance of the proposed bidirectional front-end. The values will be updated after final top-level EM extraction and plot selection.

Table 4.1: Preliminary performance summary of the proposed bidirectional Ka-band front-end.

Parameter	Value
Technology	65-nm CMOS
Frequency range	24–30 GHz
RX gain	up to 24 dB
TX gain	up to 28 dB
RX NF / NFmin	4.4–5.5 dB / 4.2–5.0 dB
Peak OP1dB	$\sim$10 dBm
Peak Psat	$\sim$13 dBm
TX/RX DC power	196 / 96 mW
Core area	$\sim$0.2 mm^2

4.9 Comparison with State-of-the-Art

A state-of-the-art comparison should include recent Ka-band and 5G FR2 bidirectional front-end modules, phased-array beamformer channels, and compact PA/LNA sharing architectures. The comparison should emphasize frequency range, technology, gain, noise figure, output compression point, power consumption, and core area.

Table 4.2: Suggested format for state-of-the-art comparison.

Work	Tech.	Freq. (GHz)	Gain (dB)	NF (dB)	OP1dB (dBm)	Area (mm^2)
This work	65-nm CMOS	24–30	24/28	4.4–5.5	$\sim$10	$\sim$0.2
Pang et al.	CMOS	28	TBD	TBD	TBD	TBD
Park and Wang	CMOS	26–39	TBD	TBD	TBD	TBD
Zhang et al.	CMOS	Ka-band	TBD	TBD	TBD	TBD

4.10 Discussion

The proposed design should be discussed in terms of its strengths and limitations. The main strengths are compact area, shared signal path, full 24–30 GHz coverage, and scalability for multi-channel arrays. The main limitations may include output power compared to base-station-class beamformer channels, post-layout EM uncertainty, and the need for measurement verification.

Placeholder: Expand this discussion after the final comparison table and post-layout results are completed.

5. CONCLUSIONS AND FUTURE WORK

5.1 Summary of the Thesis

This thesis presented a compact bidirectional RF front-end module for 24–30 GHz 5G FR2 applications. The work was motivated by the need for area-efficient front-end channels in scalable millimeter-wave phased-array and MIMO systems.

5.2 Main Achievements

The main achievements include the development of a shared bidirectional architecture, an asymmetric first stage for balancing transmitter and receiver requirements, neutralized bidirectional gain stages, transformer-based matching circuits, and system-level interpretation of the design in the context of multi-channel beamformer arrays. In addition, the thesis investigates the feasibility of implementing the proposed architecture in a standard 65-nm CMOS technology by considering area, power-consumption, passive integration, and array-level scalability trade-offs.

5.3 Limitations of the Proposed Design

The proposed design is primarily evaluated through simulations. Full-chip electromagnetic extraction, packaging effects, antenna integration, and measurement validation remain necessary for final verification. In addition, the channel output power may be more suitable for compact arrays, user equipment, customer premises equipment, or small-cell applications than for high-power macro base-station arrays.

5.4 Future Research Directions

Future work may extend the proposed front-end toward a complete multi-channel beamformer integrated circuit.

5.4.1 Multi-Channel Beamformer Integration

The proposed front-end can be replicated to realize 4-, 8-, or 16-channel beamformer ICs. This requires phase shifters, attenuators, control logic, power distribution, and careful thermal design.

5.4.2 Antenna-in-Package Implementations

Antenna-in-package integration can reduce RF routing loss between the beamformer IC and antenna elements. This direction is especially important for practical 5G FR2 modules.

5.4.3 Extension to D-Band Systems

The same system-to-circuit design methodology can be extended to D-band and sub-THz systems, although output power, noise figure, and packaging become more challenging.

5.4.4 Fabrication and Measurement Verification

Fabrication and on-wafer measurement are required to validate the proposed architecture. Future measurements should include S-parameters, noise figure, compression, PAE, stability, and large-signal behavior across the 24–30 GHz band.

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APPENDICES

The appendix provides the detailed derivation of the transformer-based matching equations used in Chapter 2.

TRANSFORMER MATCHING DERIVATIONS

This appendix provides the mathematical derivation of the transformer-based matching formulation summarized in Chapter 2. The derivation follows the notation of Bloch and Socher [12]. The purpose of this appendix is to keep the main chapter readable while still providing the complete analytical background for the normalized transformer synthesis procedure.

A.1 Coupled-inductor model

A first-order transformer is represented by two coupled inductors, L_1 and L_2, with magnetic coupling coefficient k. The physical range of the coupling coefficient is $0 < k < 1$. The equivalent circuit separates the transformer into leakage-related inductive terms and an ideal transformer. The auxiliary parameters are defined as

$$\alpha = \frac{1-k^2}{k^2}, \tag{A.1}$$

$$L = k^2 L_1. \tag{A.2}$$

The parameter α increases as the magnetic coupling becomes weaker. Therefore, it represents the nonideal leakage contribution of the transformer in the equivalent model. When k approaches unity, α tends to zero and the transformer approaches the ideal case.

A.2 Source and load impedance notation

The formulation assumes capacitive source and load impedances. This assumption is practical for CMOS RF stages because active-device input and output impedances often include significant parasitic capacitances. The impedances are written as

$$Z_S = R_S - jQ_S R_S, \tag{A.3}$$

$$Z_L = R_L - jQ_L R_L. \tag{A.4}$$

Here, $R_S = \mathrm{Re}\{Z_S\}$ and $R_L = \mathrm{Re}\{Z_L\}$ are positive real parts, while $Q_S > 0$ and $Q_L > 0$ are the capacitive impedance quality factors. The ideal transformer reflects the load impedance by the square of the turns ratio and preserves its quality factor:

$$Z_2 = N^2 Z_L = N^2 R_L - jN^2 Q_L R_L = R_2 - jQ_L R_2. \tag{A.5}$$

The corresponding admittance is

$$Y_2 = \frac{1}{Z_2} = \frac{1}{R_2}\frac{1+jQ_L}{1+Q_L^2} = G_2 + jQ_L G_2, \tag{A.6}$$

where

$$G_2 = \frac{1}{R_2(1+Q_L^2)}. \tag{A.7}$$

A.3 Intermediate matching condition

The goal of the matching circuit is to make the impedance seen by the source equal to the complex conjugate of Z_S. In the equivalent network, the intermediate impedance can be written from the source side as

$$Z_i = Z_S^* - j\omega\alpha L = R_S + jQ_S R_S - j\omega\alpha L. \tag{A.8}$$

The same intermediate node can be described from the load side in admittance form as

$$Y_i = Y_2 + \frac{1}{j\omega L}. \tag{A.9}$$

The matching condition is then

$$Z_i = \frac{1}{Y_i}. \tag{A.10}$$

Substituting (A.3)–(A.6) into (A.7), and equating the real and imaginary parts, gives

$$G_2 = \frac{R_S}{R_S^2 + (Q_S R_S - \omega\alpha L)^2}, \tag{A.11}$$

$$Q_L G_2 - \frac{1}{\omega L} = \frac{-Q_S R_S + \omega\alpha L}{R_S^2 + (Q_S R_S - \omega\alpha L)^2}. \tag{A.12}$$

A.4 Derivation of the normalized matching equation

The normalized impedance variable is defined as

$$\tilde{Z} = \frac{\omega\alpha L}{R_S}. \tag{A.13}$$

Using this definition, the term $Q_S R_S - \omega\alpha L$ becomes $R_S(Q_S - \tilde{Z})$. Substitution into the real-part condition gives

$$G_2 = \frac{1}{R_S\left[1 + \left(Q_S - \tilde{Z}\right)^2\right]}. \tag{A.14}$$

Substituting (A.11) into the imaginary-part condition and using $\omega L/R_S = \tilde{Z}/\alpha$ gives the normalized matching equation

$$\frac{Q_L}{1 + \left(Q_S - \tilde{Z}\right)^2} - \frac{\alpha}{\tilde{Z}} = \frac{-\left(Q_S - \tilde{Z}\right)}{1 + \left(Q_S - \tilde{Z}\right)^2}. \tag{A.15}$$

This equation is the central result of the derivation. It shows that the normalized transformer solution is determined only by Q_S, Q_L, and the coupling coefficient through α. The absolute impedance levels and the operating frequency scale the final inductance values but do not change the normalized solution.

A.5 Solution of the normalized impedance variable

Solving (A.12) with respect to $\tilde{Z}$ gives two possible mathematical solutions:

$$\tilde{Z}_{1,2} = \frac{2\alpha Q_S + Q_S + Q_L}{2(\alpha+1)} \pm \frac{\sqrt{(2\alpha Q_S + Q_S + Q_L)^2 - 4(\alpha^2+\alpha)(1+Q_S^2)}}{2(\alpha+1)}. \quad \text{(A.16)}$$

In practical integrated transformer design, the physically useful solution is the one that leads to reasonable inductance values, feasible winding ratios, high self-resonance frequency, and acceptable insertion loss. The alternative solution may mathematically satisfy the matching equation but can require excessively large inductors or impractical winding ratios.

A.6 Derivation of cross-quality factors

The transformer windings are expressed using the cross-quality factors

$$Q_{XL1} = \frac{\omega L_1}{R_S}, \quad \text{(A.17)}$$

$$Q_{XL2} = \frac{\omega L_2}{R_L}. \quad \text{(A.18)}$$

Using $L = k^2 L_1$ and $\alpha = (1-k^2)/k^2$, the normalized variable can be written as

$$\tilde{Z} = \frac{\omega \alpha k^2 L_1}{R_S} = (1-k^2)\frac{\omega L_1}{R_S}. \quad \text{(A.19)}$$

Therefore,

$$Q_{XL1} = \frac{\tilde{Z}}{1-k^2}. \quad \text{(A.20)}$$

The second winding value is obtained by combining the load-side reflected resistance relation with the intermediate-node quality factor. The resulting expression is

$$Q_{XL2} = \frac{\tilde{Z}}{\alpha}\frac{1+Q_L^2}{1+\left(Q_S-\tilde{Z}\right)^2}. \quad \text{(A.21)}$$

After Q_{XL1} and Q_{XL2} are found, the physical inductance values are calculated as

$$L_1 = \frac{Q_{XL1} R_S}{\omega}, \quad \text{(A.22)}$$

$$L_2 = \frac{Q_{XL2} R_L}{\omega}. \quad \text{(A.23)}$$

A.7 Finite-Q iterative correction

The derivation above assumes ideal inductors. In a practical CMOS transformer, metal resistance, skin effect, proximity effect, and substrate loss introduce finite winding quality factors. A finite inductor quality factor can be approximated by adding an effective series resistance to each winding:

$$R_{L1} = \frac{\omega L_1}{Q_{L1}}, \tag{A.24}$$

$$R_{L2} = \frac{\omega L_2}{Q_{L2}}. \tag{A.25}$$

These resistances modify the effective real parts of the source and load impedances used in the normalized synthesis. Therefore, a practical design flow can be written as follows. First, the ideal-inductor formulation is used to obtain Q_{XL1}, Q_{XL2}, L_1, and L_2. Second, expected winding quality factors are used to estimate R_{L1} and R_{L2}. Third, the effective real parts of the impedances are updated by including these parasitic resistances. Finally, the normalized synthesis is repeated until the inductance values converge.

This finite-Q correction is not intended to replace electromagnetic simulation. Instead, it improves the analytical starting point by accounting for the fact that on-chip transformer windings have finite loss. The final transformer used in the RF front-end must still be validated by EM simulation for coupling coefficient, winding quality factor, self-resonance frequency, insertion loss, and port matching.

CURRICULUM VITAE

Name Surname: Hüseyin Cahit Erbil

Education:

- B.Sc., Electronics and Communication Engineering, Istanbul Technical University, 2023.
- M.Sc., Electronics Engineering, Istanbul Technical University, expected 2026.

Professional Experience:

- RFIC Design Engineer, 2023–2026.